

LEVEL 5 · CHALLENGE

Parking Orbit

The last page of Francis Benoni's notebook is a tangle of orbital equations, and taped beneath them a printout marked only: "PARKING ORBIT RELAY — intercepted. it holds its station over one fixed point on the turning Earth, never rising, never setting. the key is where it sits above the ground."

The relay locked its final message to a single number — a whole number of kilometres you can work out from where it sits. Find it, open the lock, and hear what it was saying. Use exactly these values:

G	$6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$
Mass of Earth	$5.972 \times 10^{24} \text{ kg}$
Sidereal day	86164 s
Radius of Earth	6371 km

Files	transmission.txt, lock.py
Needs	Python 3
Flag format	CTC{...}